

Preprocessing Plan

This pipeline prepares multimodal time-series datasets (HAR, EEG, ECG) for downstream self-supervised learning by producing harmonised, reproducible outputs while preserving signal fidelity, metadata, and data provenance.

Channel Schema

For HAR (PAMAP2, WISDM), a unified six-channel representation is used: tri-axial accelerometer and tri-axial gyroscope from wrist/watch sensors, ensuring consistent sensor placement across datasets.

For EEG (EEGMMIDB), all available channels are retained together with event annotations.

For ECG (PTB-XL), all 12 leads are preserved to maintain clinical completeness.

Sampling Rate

HAR signals are resampled to **20 Hz** to harmonise PAMAP2 (100 Hz) and WISDM (20 Hz).

EEG is kept at its native **160 Hz** to preserve temporal resolution for event-based analysis.

ECG is processed at **100 Hz**, balancing computational efficiency and signal quality.

Windowing and Splitting

HAR: >> 10-second windows (no overlap, no labels) for pretraining

>> 5-second windows (50% overlap) with labels for supervised evaluation

EEG: >> Fixed 4-second windows aligned to **T1/T2 event onset** (motor imagery runs 4, 8, 12)

ECG: >> Each 10-second record is used directly without additional windowing

Subject and patient identifiers are preserved across all modalities to enable leakage-free train/validation/test splitting.

Handling Missing Values and Labels

Invalid or missing signal values are removed during preprocessing.

For HAR, transient/null labels (e.g., class 0 in PAMAP2) are excluded from supervised outputs, and dataset-specific labels are mapped to a unified schema. WISDM activities not compatible with the shared schema are excluded.

EEG uses annotation events directly, excluding incomplete windows.

ECG records with invalid or missing metadata are skipped.

Resource Considerations

Processing is performed sequentially (per subject, run, or record) to limit memory usage. Large datasets are handled in a streaming manner without loading full datasets into memory. Only representative sample packs are generated for submission to minimise storage requirements.

Engineering Risks

Key risks include cross-dataset inconsistencies, large file handling, and correct temporal alignment of windows. These are mitigated through a unified channel schema, controlled resampling, event-based windowing, and validation checks ensuring consistency, integrity, and reproducibility.